



MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
Hospital Queue Simulation and Modelling System
By

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REGISTRATION NUMBER: CIT-223-050/2022

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DECLARATION

I hereby declare that this Project is my work and has, to the best of my knowledge, not been submitted to any other institution of higher learning.

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This project has been submitted as a partial fulfillment of requirements for the Bachelor of Science in Software Engineering of Multimedia University of Kenya with my approval as the University supervisor.

Supervisor: _____

Signature: Date:

ACKNOWLEDGMENTS

I would like to acknowledge our supervisor for the support she gave us by introducing us to project design and with the exposure she gave us. I also appreciate the support of my guardians particularly financially as they ensured we had the necessary tools required to facilitate the development of the system. Thanks also goes to the faculty for ensuring we have time allocated in the school calendar to work on the project and providing the internet that facilitated research on the project. Lastly, I appreciate my fellow course-mates who provided feedback through out the development of the system. I was able to have a better understanding on what I was working on based on their ideas and I am grateful to them.

ABSTRACT

The Hospital Queue Simulation and Modelling System (HQSMS) is developed to address inefficiencies in outpatient service delivery within public hospitals, which are often characterized by long waiting times, congestion and suboptimal resource utilization. This project aims to analyze and improve patient flow by modelling and simulating hospital queue systems using a data-driven approach. The system enables healthcare administrators to evaluate service performance and test alternative operational strategies without disrupting real hospital operations.

Using a simulation-based methodology, the system is designed to model patient arrivals, service processes and queue behavior across key outpatient service points. The simulation allows for the analysis of waiting times, queue lengths and resource utilization under varying conditions. Testing and validation results demonstrated the system's effectiveness in identifying bottlenecks, predicting congestion and evaluating what-if scenarios such as changes in staffing levels and patient arrival rates.

Results indicate that the proposed system significantly improves decision-making by providing predictive insights that are not available in traditional queue management systems. Feedback from simulated scenarios showed potential reductions in patient waiting time and improved service efficiency. Future enhancements may include real-time data integration, appointment scheduling support, and AI-based demand forecasting. The system provides a practical and scalable tool for optimizing healthcare service delivery in public hospitals.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Public hospitals in Kenya experience high patient volumes, especially in outpatient departments. Patients often spend long hours waiting for consultation, laboratory services, and pharmacy services. These delays are caused by limited medical staff, inefficient patient flow, and lack of proper scheduling mechanisms.

Simulation and modelling techniques provide a way to represent real-world hospital operations using mathematical and computational models. By simulating patient arrival rates, service times, and resource availability, it becomes possible to evaluate hospital performance without disrupting real operations. This study focuses on using simulation to analyse queue behaviour and improve service delivery in Kenyan public hospitals.

1.2 Problem Statement

Long patient waiting times in public hospitals negatively affect quality of healthcare, patient satisfaction, and staff productivity. Current queue management practices are mostly manual and do not provide hospital administrators with predictive insights for decision-making. Without proper modelling, it is difficult to determine optimal staffing levels or service arrangements. There is therefore a need for a simulation-based approach to analyse and improve queue management in public hospitals.

1.3 Aim of the Study

The aim of this project is to develop a simulation model that analyses patient queues in public hospitals and proposes improvements to reduce waiting time and congestion.

1.4 General Objective

To model and simulate outpatient queue systems in Kenyan public hospitals in order to enhance service efficiency.

1.5 Specific Objectives

- To study the current patient flow and queuing system in outpatient departments.
- To develop a simulation model representing patient arrivals and service processes.
- To analyse waiting times, queue lengths, and server utilisation.

- To evaluate alternative scenarios for improving service efficiency.

1.6 Justification of the Study

This study will help hospital administrators understand bottlenecks in service delivery without interfering with real hospital operations. The simulation model will support data-driven decision-making, improve patient satisfaction, and optimize the use of limited healthcare resources. The project is also academically relevant as it applies simulation and modelling concepts taught in computer science.

1.7 Scope of the Study

The study will focus on outpatient services only. The simulation will model patient arrival, registration, consultation, and pharmacy stages. Inpatient services and emergency units are excluded. Data will be collected from one selected public hospital in Kenya.

1.8 Assumptions

- Patient arrivals follow a known probability distribution.
- Service times are independent for each patient.
- Hospital resources remain constant during simulation runs.
- Patients follow a first-come-first-served discipline.

1.9 Limitations

- Limited access to real hospital data may affect model accuracy.
- Human factors such as staff behaviour changes are difficult to model.
- The simulation results may not fully generalize to all hospitals.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Simulation and modelling have been widely used in healthcare systems to improve operational efficiency. This chapter reviews existing studies and systems related to hospital queue management.

2.2 Hospital Queue Simulation Systems

Several researchers have applied discrete event simulation to hospital settings to study patient flow, resource allocation, and service times. These models help predict congestion and evaluate improvement strategies.

2.3 Limitations of Existing Studies.

Most existing models are developed for hospitals in developed countries and may not reflect constraints faced by Kenyan public hospitals such as understaffing and high patient influx

2.4 Research Gap

There is limited localized research focusing on simulation of outpatient queues in Kenyan public hospitals. This study addresses this gap by developing a context-specific model.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter describes the methodology used to develop the simulation model for analysing queue management in public hospitals.

3.2 Research Design

The study will adopt a simulation-based research design using discrete event simulation to represent patient flow and service processes.

3.3 Simulation Approach

Discrete event simulation will be used to model patient arrivals, service points, waiting queues, and service completion events.

3.4 Data Collection Methods

- ☐ **Observation:** Studying patient flow within outpatient departments.
- ☐ **Interviews:** Discussions with hospital staff on service procedures.
- ☐ **Document Review:** Reviewing hospital records and reports.

3.5 Tools and Technologies

- **Simulation Tool:** Python (SimPy) or AnyLogic
- **Data Analysis:** Python/Excel
- **Documentation:** Microsoft Word

3.6 Model Validation

The simulation results will be validated by comparing model outputs with observed hospital performance indicators.

3.7 Ethical Considerations

Patient data will be anonymized and used strictly for academic purposes.